

ELITE MATHEMATICS

Pearson Edexcel International A Level

# Mechanics 1

## Adaptive Classified

Chapter 07 • Newton's Second Law

### Questions

Student book • questions with writing space

- 80 pages
- Past papers up to June 2026

## MECHANICS 1 | TOPIC 07

# Newton's Second Law

Your classified route

**25**

QUESTIONS

**287**

MARKS

**4**

FAMILIES

**10**

VARIANTS

DR ESLAM AHMED

**CLASSIFIED BY IDEA**

Read the diagram, choose the model, then use the working space to complete the mechanics.

## TOPIC 07 | FAMILY 01

# Newton's second law for one particle

**WHAT THIS FAMILY ASKS**

Resolve the resultant force and use  $F = ma$  for a single particle, vehicle or lift.

**VARIANT ROUTE****V01**    Scalar  $F = ma$ **V02**    Lift or apparent reaction**6 classified questions**

44 total marks

3. Two students observe a book of mass  $0.2\text{ kg}$  fall vertically from rest from a shelf that is  $1.5\text{ m}$  above the floor.

Student  $A$  suggests that the book is modelled as a particle falling freely under gravity.

- (a) Use student  $A$ 's model to find the time taken for the book to reach the floor.

(3)

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3. Two students observe a book of mass  $0.2\text{ kg}$  fall vertically from rest from a shelf that is  $1.5\text{ m}$  above the floor.

Student  $A$  suggests that the book is modelled as a particle falling freely under gravity.

Student  $B$  suggests an improved model where the book is modelled as a particle experiencing a constant resistance to motion of magnitude  $R$  newtons.

Given that the time taken for the book to reach the floor is  $0.6$  seconds,

(b) use student  $B$ 's model to find the value of  $R$

(5)

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3. A hammer is used to hit a tent peg into soft ground.

The hammer has mass 1.8 kg and the tent peg has mass 0.2 kg.

The hammer and tent peg are both modelled as particles and the impact is modelled as a direct collision.

Immediately before the impact, the tent peg is stationary and the hammer is moving vertically downwards with speed  $10 \text{ m s}^{-1}$

Immediately after the impact, the hammer and tent peg move together, vertically downwards, with the **same** speed  $v \text{ m s}^{-1}$

- (a) Find the value of  $v$

(2)

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Immediately before the impact, the tent peg is stationary and the hammer is moving vertically downwards with speed  $10 \text{ m s}^{-1}$

Immediately after the impact, the hammer and tent peg move together, vertically downwards, with the **same** speed  $v \text{ m s}^{-1}$

- (b) Find the magnitude of the impulse exerted on the tent peg by the hammer, stating the units of your answer.

(3)

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3. A hammer is used to hit a tent peg into soft ground.

The hammer has mass 1.8 kg and the tent peg has mass 0.2 kg.

Immediately after the impact, the hammer and tent peg move together, vertically downwards, with the **same** speed  $v \text{ m s}^{-1}$

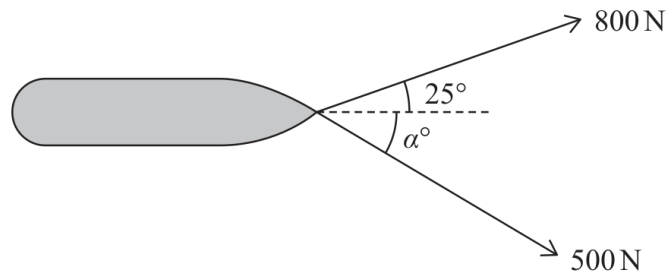
The ground exerts a constant vertical resistive force of magnitude  $R$  newtons, bringing the hammer and tent peg to rest after they travel a distance of 12 cm.

- (c) Find the value of  $R$ .

(5)

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4.

**Figure 2**

Two ropes are attached to a point on the front of a barge.  
The barge is being pulled horizontally in a straight line along the centre of a long straight canal.

One rope makes an angle of  $25^\circ$  with the direction of motion of the barge and has a tension of 800 N.

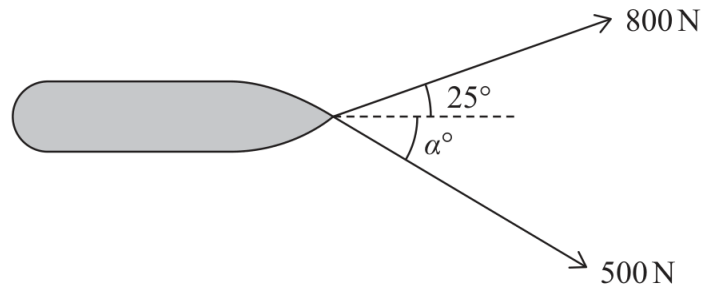
The other rope makes an angle of  $\alpha^\circ$  with the direction of motion of the barge and has a tension of 500 N, as shown in Figure 2.

Both ropes are horizontal.

(a) Find the value of  $\alpha$

(3)

4.

**Figure 2**

One rope makes an angle of  $25^\circ$  with the direction of motion of the barge and has a tension of 800 N.

The other rope makes an angle of  $\alpha^\circ$  with the direction of motion of the barge and has a tension of 500 N, as shown in Figure 2.

Both ropes are horizontal.

The mass of the barge is 15 tonnes and the resistance to the motion of the barge is a constant force of magnitude 750 N.

(b) Find the acceleration of the barge.

**(4)**

4.

Figure 1 shows a large bucket used by a crane on a building site to move materials between the ground and the top of the building. The mass of the bucket is 15 kg.

The bucket is attached to a vertical cable with the bottom of the bucket horizontal. The cable is modelled as light and inextensible.

When the bucket is on the ground, a bag of cement of mass 25 kg is placed in the bucket.

The bucket with the bag of cement moves vertically upwards with constant acceleration  $0.2 \text{ ms}^{-2}$ . Air resistance is modelled as being negligible.

(a) Find the tension in the cable.

(3)

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4.

Figure 1 shows a large bucket used by a crane on a building site to move materials between the ground and the top of the building. The mass of the bucket is 15 kg.

The bucket is attached to a vertical cable with the bottom of the bucket horizontal. The cable is modelled as light and inextensible.

At the top of the building, the bag of cement is removed. A box of tools of mass 12 kg is now placed in the bucket.

Later on the bucket with the box of tools is moving vertically downwards with constant deceleration  $0.1 \text{ ms}^{-2}$ . Air resistance is again modelled as being negligible.

(b) Find the magnitude of the normal reaction between the bucket and the box of tools.

(3)

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4.

Two children, Alan and Bhavana, are standing on the horizontal floor of a lift, as shown in Figure 3.

The lift has mass 250 kg. The lift is raised vertically upwards with constant acceleration by a vertical cable which is attached to the top of the lift. The cable is modelled as being light and inextensible. While the lift is accelerating upwards, the tension in the cable is 3616 N.

As the lift accelerates upwards, the floor of the lift exerts a force of magnitude 565 N on Alan and a force of magnitude 226 N on Bhavana.

Air resistance is modelled as being negligible and Alan and Bhavana are modelled as particles.

(a) By considering the forces acting on the lift only, find the acceleration of the lift.

(3)

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4.

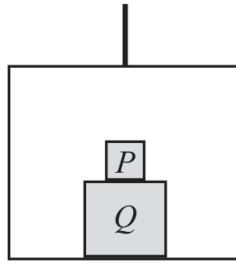
As the lift accelerates upwards, the floor of the lift exerts a force of magnitude  $565\text{ N}$  on Alan and a force of magnitude  $226\text{ N}$  on Bhavana.

Air resistance is modelled as being negligible and Alan and Bhavana are modelled as particles.

(b) Find the mass of Alan.

(3)

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**Figure 4**

A simple lift operates by means of a vertical cable which is attached to the top of the lift.

The lift has mass  $m$

A box  $Q$  is placed on the floor of the lift.

A box  $P$  is placed directly on top of box  $Q$ , as shown in Figure 4.

The tension in the cable is  $\frac{42mg}{5}$

The lift and its contents move vertically upwards with acceleration  $\frac{2g}{5}$

Using the model,

(a) find, in terms of  $m$ , the combined mass of boxes  $P$  and  $Q$

(4)

A simple lift operates by means of a vertical cable which is attached to the top of the lift.

The lift has mass  $m$

A box  $Q$  is placed on the floor of the lift.

A box  $P$  is placed directly on top of box  $Q$ , as shown in Figure 4.

The lift and its contents move vertically upwards with acceleration  $\frac{2g}{5}$

Using the model,

During the motion of the lift, the force exerted on box  $P$  by box  $Q$  is  $\frac{14mg}{5}$

Using the model,

(b) find, in terms of  $m$ , the mass of box  $P$

(3)

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## TOPIC 07 | FAMILY 02

# Connected-particle systems

**WHAT THIS FAMILY ASKS**

Model particles linked by light inextensible strings over smooth pulleys and solve coupled equations.

**VARIANT ROUTE**

**V01** Horizontal-table particle connected to a hanging particle

**V02** Incline connected to another particle

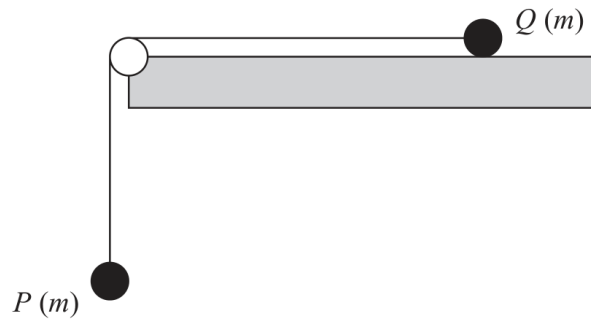
**V03** Multiple strings or tension levels

**V04** Two vertically hanging connected particles

**V05** Limiting threshold followed by connected motion

**13 classified questions**

178 total marks

**Figure 4**

A particle  $P$  of mass  $m$  is attached to one end of a light inextensible string. Another particle  $Q$ , also of mass  $m$ , is attached to the other end of the string. The string passes over a small smooth pulley which is fixed at the edge of a rough horizontal table. Particle  $Q$  is held at rest on the table and particle  $P$  hangs vertically below the pulley with the string taut, as shown in Figure 4.

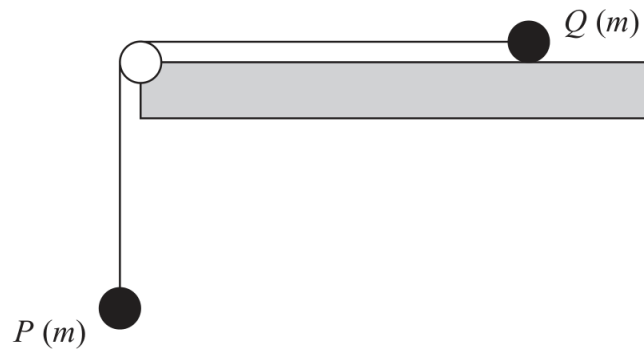
The coefficient of friction between  $Q$  and the table is  $\mu$ , where  $\mu < 1$

Particle  $Q$  is released from rest.

The tension in the string before  $Q$  hits the pulley is  $kmg$ , where  $k$  is a constant.

(a) Find  $k$  in terms of  $\mu$ .

(7)

**Figure 4**

The coefficient of friction between  $Q$  and the table is  $\mu$ , where  $\mu < 1$

Particle  $Q$  is released from rest.

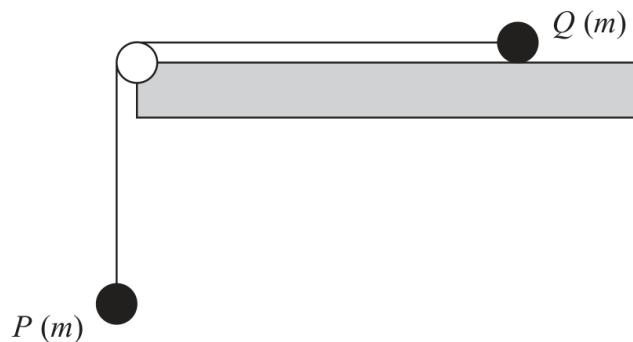
The tension in the string before  $Q$  hits the pulley is  $kmg$ , where  $k$  is a constant.

Given that  $Q$  is initially a distance  $d$  from the pulley,

(b) find, in terms of  $d$ ,  $g$  and  $\mu$ , the time taken by  $Q$ , after release, to reach the pulley.

(4)

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**Figure 4**

A particle  $P$  of mass  $m$  is attached to one end of a light inextensible string. Another particle  $Q$ , also of mass  $m$ , is attached to the other end of the string. The string passes over a small smooth pulley which is fixed at the edge of a rough horizontal table. Particle  $Q$  is held at rest on the table and particle  $P$  hangs vertically below the pulley with the string taut, as shown in Figure 4.

The coefficient of friction between  $Q$  and the table is  $\mu$ , where  $\mu < 1$

Particle  $Q$  is released from rest.

(c) Describe what would happen if  $\mu \geq 1$ , giving a reason for your answer.

(2)

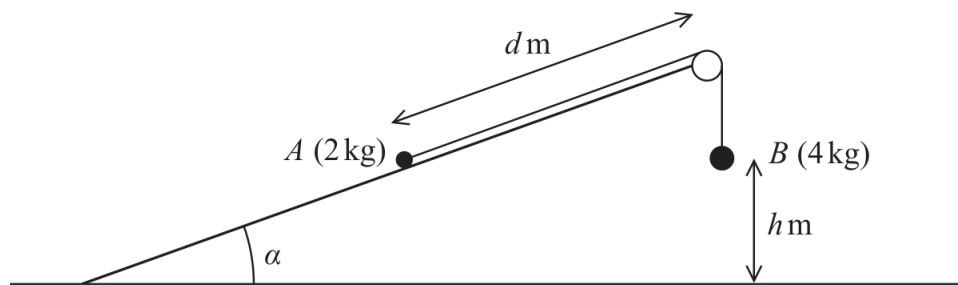


Figure 3

at the top of a rough plane. The plane is inclined to the horizontal ground at an angle  $\alpha$  where  $\tan \alpha = \frac{3}{4}$ . The particle  $A$  is held at rest on the plane at a distance  $d$  metres from the pulley. The particle  $B$  hangs freely at rest, vertically below the pulley, at a distance  $h$  metres above the ground, as shown in Figure 3. The part of the string between  $A$  and the pulley is parallel to a line of greatest slope of the plane. The coefficient of friction between  $A$  and the plane is  $\frac{1}{4}$ .

The system is released from rest with the string taut and  $B$  descends.

- (a) Find the tension in the string as  $B$  descends.

(9)

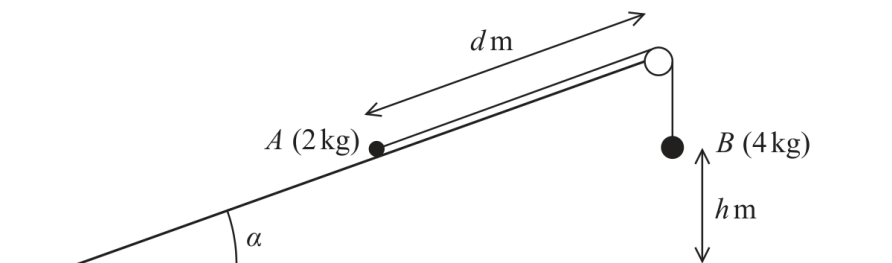


Figure 3

at the top of a rough plane. The plane is inclined to the horizontal ground at an angle  $\alpha$  where  $\tan \alpha = \frac{3}{4}$ . The particle  $A$  is held at rest on the plane at a distance  $d$  metres from the pulley. The particle  $B$  hangs freely at rest, vertically below the pulley, at a distance  $h$  metres above the ground, as shown in Figure 3. The part of the string between  $A$  and the pulley is parallel to a line of greatest slope of the plane. The coefficient of friction between  $A$  and the plane is  $\frac{1}{4}$ .

On hitting the ground,  $B$  immediately comes to rest.

Given that  $A$  comes to rest before reaching the pulley,

(b) find, in terms of  $h$ , the range of possible values of  $d$ .

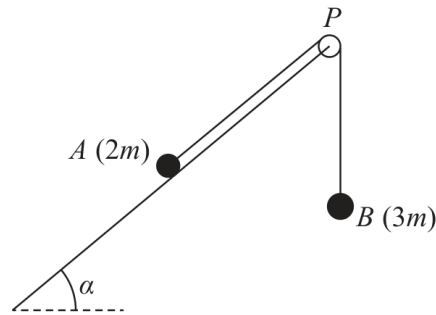
(7)

Two particles,  $A$  and  $B$ , have masses  $2\text{ kg}$  and  $4\text{ kg}$  respectively. The particles are connected by a light inextensible string. The string passes over a small smooth pulley which is fixed at the top of a rough plane. The plane is inclined to the horizontal ground at an angle  $\alpha$  where  $\tan \alpha = \frac{3}{4}$ . The particle  $A$  is held at rest on the plane at a distance  $d$  metres from the pulley. The particle  $B$  hangs freely at rest, vertically below the pulley, at a distance  $h$  metres above the ground, as shown in Figure 3. The part of the string between  $A$  and the pulley is parallel to a line of greatest slope of the plane. The coefficient of friction between  $A$  and the plane is  $\frac{1}{4}$ .

- (c) State one physical factor, other than air resistance, that could be taken into account to make the model described above more realistic.

(1)

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**Figure 4**

One end of a light inextensible string is attached to a particle  $A$  of mass  $2m$ . The other end of the string is attached to a particle  $B$  of mass  $3m$ . The string passes over a small, smooth, light pulley  $P$  which is fixed at the top of a rough inclined plane. The plane is inclined to

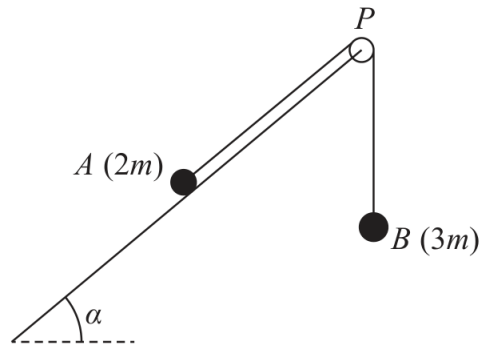
The coefficient of friction between  $A$  and the plane is  $\frac{1}{2}$

Particle  $A$  is released and begins to move up the plane.

For the motion before  $A$  reaches the pulley,

- (a) (i) write down an equation of motion for  $A$ ,  
 (ii) write down an equation of motion for  $B$ ,

(4)

**Figure 4**

The coefficient of friction between  $A$  and the plane is  $\frac{1}{2}$

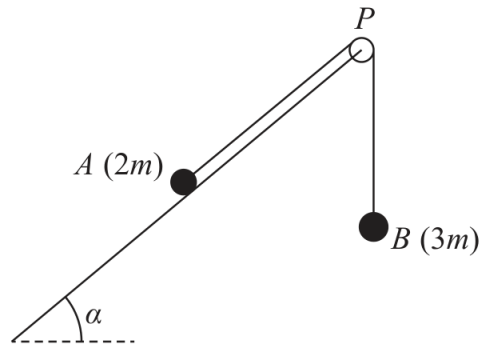
Particle  $A$  is released and begins to move up the plane.

For the motion before  $A$  reaches the pulley,

(b) find, in terms of  $g$ , the acceleration of  $A$ ,

(5)

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**Figure 4**

One end of a light inextensible string is attached to a particle  $A$  of mass  $2m$ . The other end of the string is attached to a particle  $B$  of mass  $3m$ . The string passes over a small, smooth, light pulley  $P$  which is fixed at the top of a rough inclined plane. The plane is inclined to the horizontal at an angle  $\alpha$ , where  $\tan \alpha = \frac{3}{4}$

(c) find the magnitude of the force exerted on the pulley by the string.

(4)

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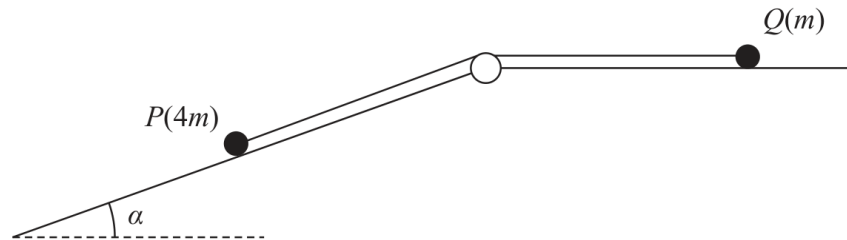
One end of a light inextensible string is attached to a particle  $A$  of mass  $2m$ . The other end of the string is attached to a particle  $B$  of mass  $3m$ . The string passes over a small, smooth, light pulley  $P$  which is fixed at the top of a rough inclined plane. The plane is inclined to

(d) State how you have used the information that  $P$  is a smooth pulley.

(1)

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1.

**Figure 4**

The particle  $P$  is attached to one end of a light inextensible string.

The string passes over a small smooth pulley that is fixed at the top of the plane. The other end of the string is attached to a particle  $Q$  of mass  $m$  which lies on a smooth horizontal plane.

The system is released from rest with the string taut, as shown in Figure 4, and  $P$  moves down the plane.

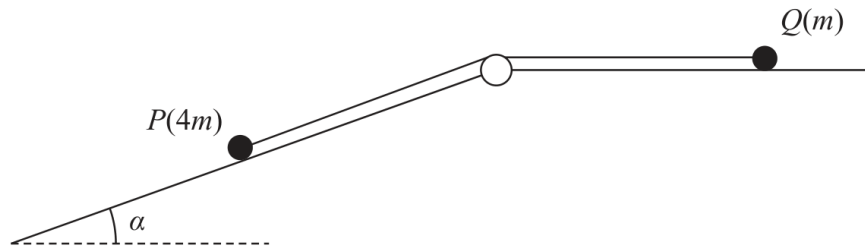
The coefficient of friction between  $P$  and the plane is  $\frac{1}{4}$

For the motion before  $Q$  reaches the pulley

(a) write down an equation of motion for  $Q$ ,

(1)

1.

**Figure 4**

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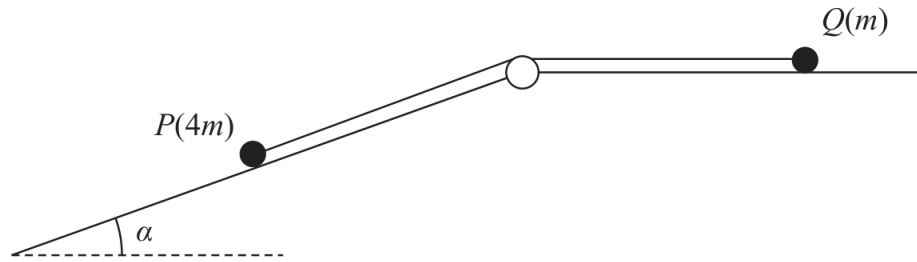
The coefficient of friction between  $P$  and the plane is  $\frac{1}{4}$

For the motion before  $Q$  reaches the pulley

(b) find, in terms of  $m$  and  $g$ , the tension in the string,

(7)

1.

**Figure 4**

The particle  $P$  is attached to one end of a light inextensible string.

The string passes over a small smooth pulley that is fixed at the top of the plane. The other end of the string is attached to a particle  $Q$  of mass  $m$  which lies on a smooth horizontal plane.

The string lies along the horizontal plane and in the vertical plane that contains the pulley and a line of greatest slope of the inclined plane.

(c) find the magnitude of the force exerted on the pulley by the string.

**(4)**

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The particle  $P$  is attached to one end of a light inextensible string.

The string passes over a small smooth pulley that is fixed at the top of the plane. The other end of the string is attached to a particle  $Q$  of mass  $m$  which lies on a smooth horizontal plane.

- (d) State where in your working you have used the information that the string is light.

(1)

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Figure 4 shows a block  $A$  of mass  $m$  held at rest on a rough plane.

The plane is inclined at an angle  $\alpha$  to the horizontal and the coefficient of friction between the block and the plane is  $\mu$ .

One end of a light inextensible string is now attached to  $A$ . The string passes over a small smooth pulley which is fixed at the top of the plane.

The other end of the string is attached to a block  $B$  of mass  $km$ .

Block  $B$  hangs vertically below the pulley, with the string taut.

The string from  $A$  to the pulley lies along a line of greatest slope of the plane.

Both  $A$  and  $B$  are modelled as particles.

When the system is released from rest,  $A$  moves up the plane and the tension in the string is  $\frac{4mg}{3}$

Given that  $\mu = \frac{1}{3}$  and  $\tan \alpha = \frac{7}{24}$

(a) (i) find the magnitude of the acceleration of  $A$ , giving your answer in terms of  $g$ ,

(ii) find the value of  $k$ .

(9)

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7.

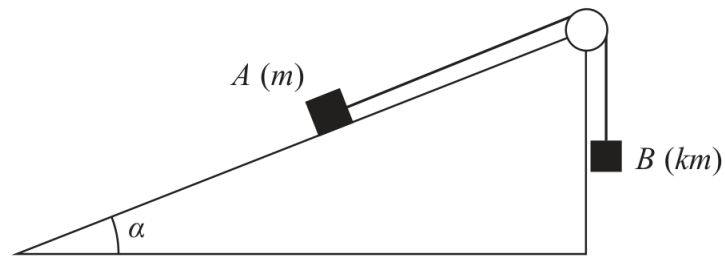


Figure 4

When the system is released from rest,  $A$  moves up the plane and the tension in the string is  $\frac{4mg}{3}$

Given that  $\mu = \frac{1}{3}$  and  $\tan \alpha = \frac{7}{24}$

- (b) Find the magnitude of the resultant force exerted on the pulley by the string, giving your answer in terms of  $m$  and  $g$ .

(4)

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One end of a light inextensible string is attached to a particle  $A$  of mass  $2m$ . The other end of the string is attached to a particle  $B$  of mass  $3m$ . Particle  $A$  is held at rest on a rough plane which is inclined to horizontal ground at an angle  $\alpha$ , where  $\tan\alpha = \frac{5}{12}$

The string passes over a small smooth pulley  $P$  which is fixed at the top of the plane. Particle  $B$  hangs vertically below  $P$  with the string taut, at a height  $h$  above the ground, as shown in Figure 4.

The part of the string between  $A$  and  $P$  lies along a line of greatest slope of the plane. The two particles, the string and the pulley all lie in the same vertical plane.

The coefficient of friction between  $A$  and the plane is  $\frac{11}{36}$

The particle  $A$  is released from rest and begins to move up the plane.

(a) Show that the frictional force acting on  $A$  as it moves up the plane is  $\frac{22mg}{39}$  (3)

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8.

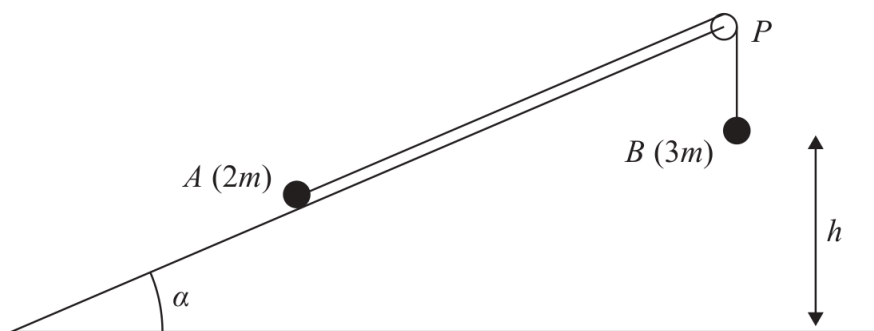


Figure 4

One end of a light inextensible string is attached to a particle  $A$  of mass  $2m$ . The other end of the string is attached to a particle  $B$  of mass  $3m$ . Particle  $A$  is held at rest on a rough plane which is inclined to horizontal ground at an angle  $\alpha$ , where  $\tan \alpha = \frac{5}{12}$ .

The string passes over a small smooth pulley  $P$  which is fixed at the top of the plane. Particle  $B$  hangs vertically below  $P$  with the string taut, at a height  $h$  above the ground, as shown in Figure 4.

(b) Write down an equation of motion for  $B$ .

(2)

One end of a light inextensible string is attached to a particle  $A$  of mass  $2m$ . The other end of the string is attached to a particle  $B$  of mass  $3m$ . Particle  $A$  is held at rest on a rough plane which is inclined to horizontal ground at an angle  $\alpha$ , where  $\tan \alpha = \frac{5}{12}$

The string passes over a small smooth pulley  $P$  which is fixed at the top of the plane. Particle  $B$  hangs vertically below  $P$  with the string taut, at a height  $h$  above the ground, as shown in Figure 4.

The part of the string between  $A$  and  $P$  lies along a line of greatest slope of the plane. The two particles, the string and the pulley all lie in the same vertical plane.

The coefficient of friction between  $A$  and the plane is  $\frac{11}{36}$

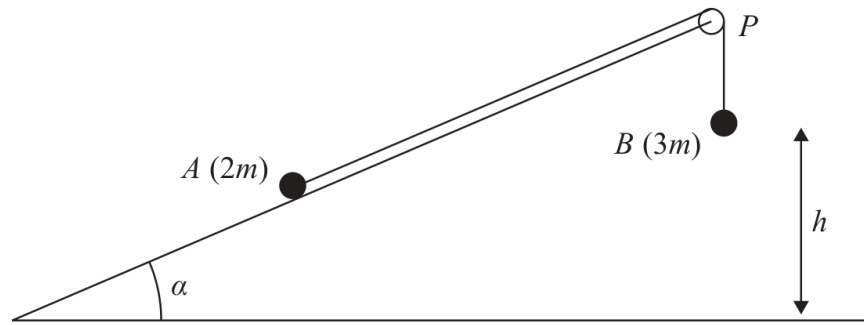
The particle  $A$  is released from rest and begins to move up the plane.

(c) Show that the acceleration of  $A$  immediately after its release is  $\frac{1}{3}g$

(4)

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8.

**Figure 4**

The part of the string between  $A$  and  $P$  lies along a line of greatest slope of the plane. The two particles, the string and the pulley all lie in the same vertical plane.

The coefficient of friction between  $A$  and the plane is  $\frac{11}{36}$

The particle  $A$  is released from rest and begins to move up the plane.

In the subsequent motion,  $A$  comes to rest before it reaches the pulley.

- (d) Find, in terms of  $h$ , the total distance travelled by  $A$  from when it was released from rest to when it first comes to rest again.

**(6)**

A block  $P$  of mass  $2m$  is held at rest on a fixed rough plane.

The plane is inclined to the horizontal at an angle  $\alpha$ , where  $\tan \alpha = \frac{5}{12}$

One end of a light inextensible string is attached to  $P$ .

The string is parallel to a line of greatest slope of the plane and passes over a smooth light pulley which is fixed at the top of the plane.

The other end of the string is attached to a block  $Q$  of mass  $5m$ .

Block  $Q$  hangs vertically below the pulley, as shown in Figure 5.

The system is released from rest with the string taut and block  $P$  moves up the plane.

Immediately after the system is released, the tension in the string is  $T$  and the acceleration of the blocks is  $a$ .

The blocks are modelled as particles and air resistance is ignored.

(a) Write down an equation of motion for  $Q$ .

(2)

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A block  $P$  of mass  $2m$  is held at rest on a fixed rough plane.

The plane is inclined to the horizontal at an angle  $\alpha$ , where  $\tan \alpha = \frac{5}{12}$

One end of a light inextensible string is attached to  $P$ .

The string is parallel to a line of greatest slope of the plane and passes over a smooth light pulley which is fixed at the top of the plane.

The other end of the string is attached to a block  $Q$  of mass  $5m$ .

Block  $Q$  hangs vertically below the pulley, as shown in Figure 5.

The system is released from rest with the string taut and block  $P$  moves up the plane.

Immediately after the system is released, the tension in the string is  $T$  and the acceleration of the blocks is  $a$ .

The blocks are modelled as particles and air resistance is ignored.

The coefficient of friction between  $P$  and the plane is  $\frac{1}{8}$

(b) Find  $T$  in terms of  $m$  and  $g$ .

(7)

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One end of a light inextensible string is attached to  $P$ .

The string is parallel to a line of greatest slope of the plane and passes over a smooth light pulley which is fixed at the top of the plane.

The other end of the string is attached to a block  $Q$  of mass  $5m$ .

Block  $Q$  hangs vertically below the pulley, as shown in Figure 5.

The system is released from rest with the string taut and block  $P$  moves up the plane.

Immediately after the system is released, the tension in the string is  $T$  and the acceleration of the blocks is  $a$ .

(c) State how the solution to part (b) uses the fact that the string is inextensible.

(1)

DR ESAM AHMED

7.

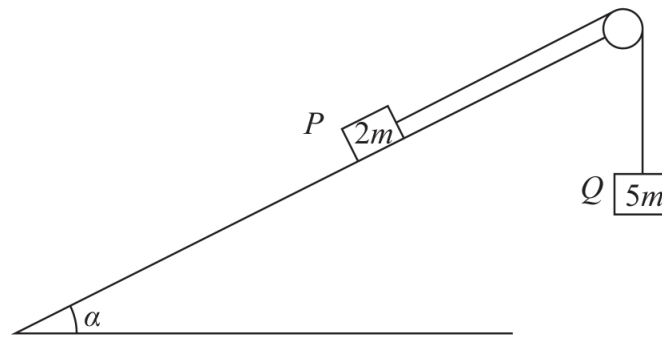


Figure 5

The plane is inclined to the horizontal at an angle  $\alpha$ , where  $\tan \alpha = \frac{5}{12}$

Immediately after the system is released, the tension in the string is  $T$  and the acceleration of the blocks is  $a$ .

The magnitude of the force exerted on the pulley by the string is  $kmg$ .

(d) Find the value of  $k$  to 3 significant figures.

(4)

One end of a light inextensible string is attached to a particle  $A$  of mass  $3m$ . The other end of the string is attached to a particle  $B$  of mass  $2m$ . Particle  $A$  lies on a **rough** plane which is inclined to the horizontal at an angle  $\alpha$ , where  $\tan \alpha = \frac{3}{4}$ . Particle  $B$  lies on a **smooth** horizontal plane. The string passes over a small smooth pulley  $P$  which is fixed at the top of the inclined plane, as shown in Figure 4.

The two particles, the string and the pulley all lie in the same vertical plane. The portion of the string between  $A$  and  $P$  lies along a line of greatest slope of the inclined plane.

The coefficient of friction between  $A$  and the inclined plane is  $\frac{1}{4}$

The system is released from rest with the string taut and  $A$  begins to move down the inclined plane.

(a) Find, in terms of  $m$  and  $g$ , the magnitude of the frictional force acting on  $A$ .

(3)

DR ESAM AHMED

8.

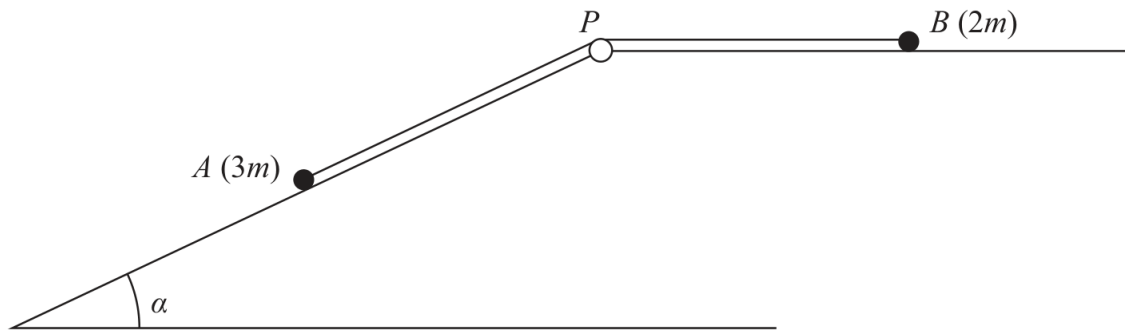


Figure 4

One end of a light inextensible string is attached to a particle  $A$  of mass  $3m$ . The other end of the string is attached to a particle  $B$  of mass  $2m$ . Particle  $A$  lies on a **rough** plane which is inclined to the horizontal at an angle  $\alpha$ , where  $\tan \alpha = \frac{3}{4}$ . Particle  $B$  lies on a **smooth** horizontal plane. The string passes over a small smooth pulley  $P$  which is fixed at the top of the inclined plane, as shown in Figure 4.

(b) Write down an equation of motion for  $B$ .

(1)

DR ESAM AHMED

One end of a light inextensible string is attached to a particle  $A$  of mass  $3m$ . The other end of the string is attached to a particle  $B$  of mass  $2m$ . Particle  $A$  lies on a **rough** plane which is inclined to the horizontal at an angle  $\alpha$ , where  $\tan \alpha = \frac{3}{4}$ . Particle  $B$  lies on a **smooth** horizontal plane. The string passes over a small smooth pulley  $P$  which is fixed at the top of the inclined plane, as shown in Figure 4.

The two particles, the string and the pulley all lie in the same vertical plane. The portion of the string between  $A$  and  $P$  lies along a line of greatest slope of the inclined plane.

The coefficient of friction between  $A$  and the inclined plane is  $\frac{1}{4}$

The system is released from rest with the string taut and  $A$  begins to move down the inclined plane.

(c) Show that the tension in the string is  $\frac{12mg}{25}$  (4)

DR ESAM AHMED

One end of a light inextensible string is attached to a particle  $A$  of mass  $3m$ . The other end of the string is attached to a particle  $B$  of mass  $2m$ . Particle  $A$  lies on a **rough** plane which is inclined to the horizontal at an angle  $\alpha$ , where  $\tan \alpha = \frac{3}{4}$ . Particle  $B$  lies on a **smooth** horizontal plane. The string passes over a small smooth pulley  $P$  which is fixed at the top of the inclined plane, as shown in Figure 4.

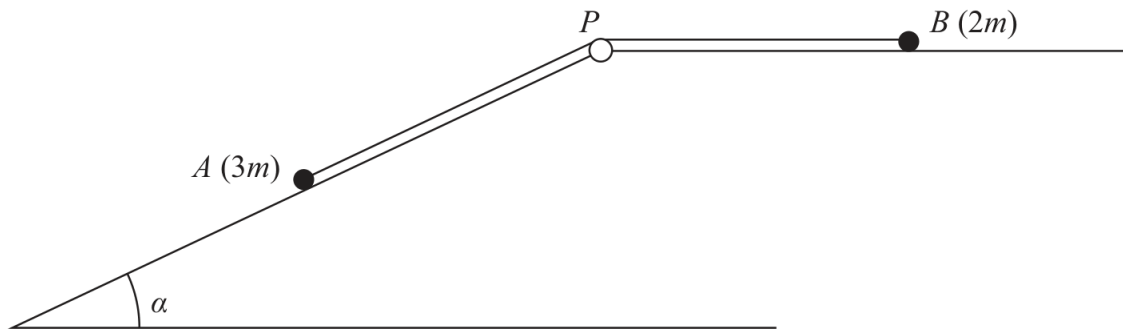
The two particles, the string and the pulley all lie in the same vertical plane. The portion of the string between  $A$  and  $P$  lies along a line of greatest slope of the inclined plane.

(d) State how you have used the fact that the string is inextensible.

(1)

DR ESAM AHMED

8.

**Figure 4**

The two particles, the string and the pulley all lie in the same vertical plane. The portion of the string between  $A$  and  $P$  lies along a line of greatest slope of the inclined plane.

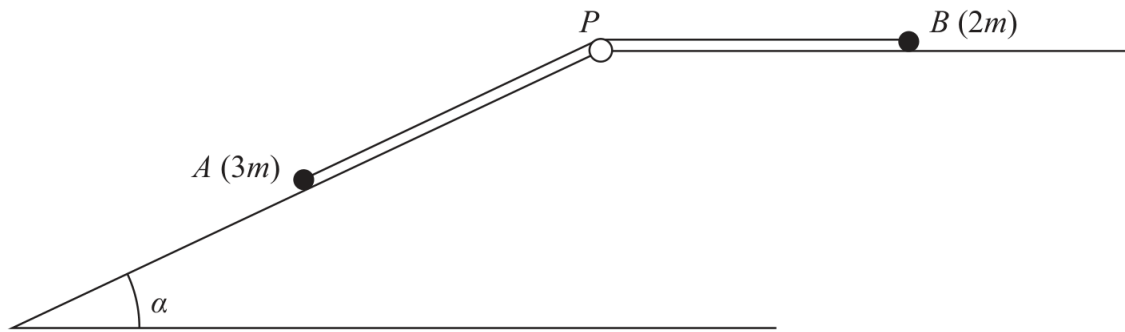
The force exerted on the pulley by the string is  $S$ .

(e) State clearly, giving your answer in terms of  $\alpha$ , the direction of  $S$ .

**(1)**

DR ESAM AHMED

8.

**Figure 4**

The two particles, the string and the pulley all lie in the same vertical plane. The portion of the string between  $A$  and  $P$  lies along a line of greatest slope of the inclined plane.

The force exerted on the pulley by the string is  $S$ .

(f) Find, in terms of  $m$  and  $g$ , the magnitude of  $S$ .

**(4)**

DR ESAM AHMED

3.

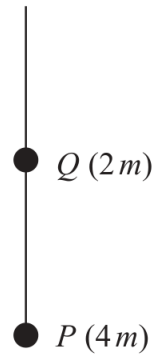


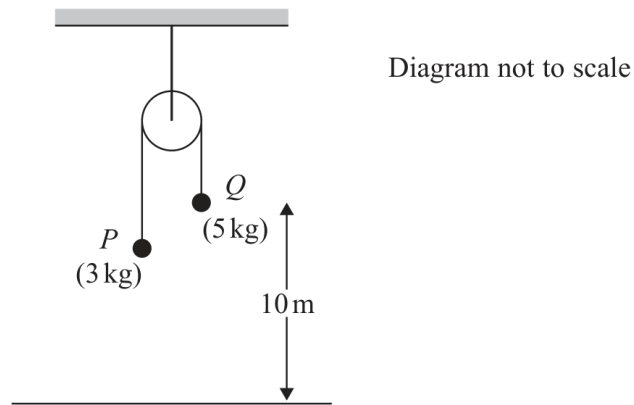
Figure 1

Two particles,  $P$  and  $Q$ , have masses  $4m$  and  $2m$  respectively. The particles are connected by a light inextensible string. A second light inextensible string has one end attached to  $Q$ . Both strings are taut and vertical, as shown in Figure 1.

The particles are **accelerating** vertically **downwards**.

Given that the tension in the string connecting the two particles is  $3mg$ , find, in terms of  $m$  and  $g$ , the tension in the upper string.

(6)

**Figure 3**

Two particles,  $P$  and  $Q$ , have masses 3 kg and 5 kg respectively. The particles are connected by a light inextensible string which passes over a small smooth fixed pulley.

The particles are released from rest with the string taut and the hanging parts of the string vertical, as shown in Figure 3.

Immediately after the particles are released from rest,  $P$  moves upwards with acceleration  $a \text{ m s}^{-2}$  and the tension in the string is  $T$  newtons.

(a) Write down an equation of motion for  $P$ .

(2)

Two particles,  $P$  and  $Q$ , have masses 3 kg and 5 kg respectively. The particles are connected by a light inextensible string which passes over a small smooth fixed pulley.

The particles are released from rest with the string taut and the hanging parts of the string vertical, as shown in Figure 3.

Immediately after the particles are released from rest,  $P$  moves upwards with acceleration  $a \text{ m s}^{-2}$  and the tension in the string is  $T$  newtons.

(b) Find the value of  $T$ .

(4)

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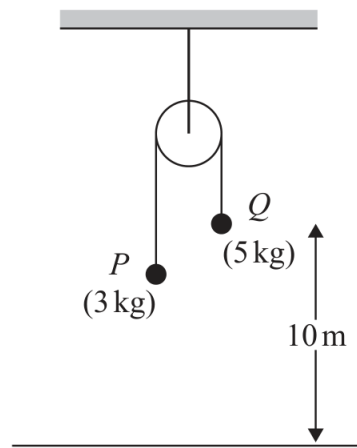


Diagram not to scale

**Figure 3**

The total force acting on the pulley due to the string has magnitude  $F$  newtons.

(c) Find the value of  $F$ .

(2)

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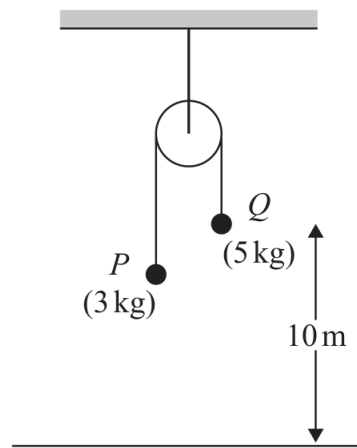


Diagram not to scale

**Figure 3**

Initially,  $Q$  is 10 m above horizontal ground and  $P$  is more than 2 m below the pulley.

At the instant when  $Q$  has descended a distance of 2 m, the string breaks and  $Q$  falls to the ground.

(d) Find the speed of  $Q$  at the instant it hits the ground.

**(5)**

DR ESAM AHMED

One end of a light inextensible string is attached to a particle  $A$  of mass  $2m$ .  
The other end is attached to a particle  $B$  of mass  $3m$ .  
The string passes over a small smooth fixed pulley.  
The string is taut and both straight parts of the string are vertical.  
Both particles are held at a distance  $0.5\text{ m}$  above a horizontal surface, as shown in Figure 4.

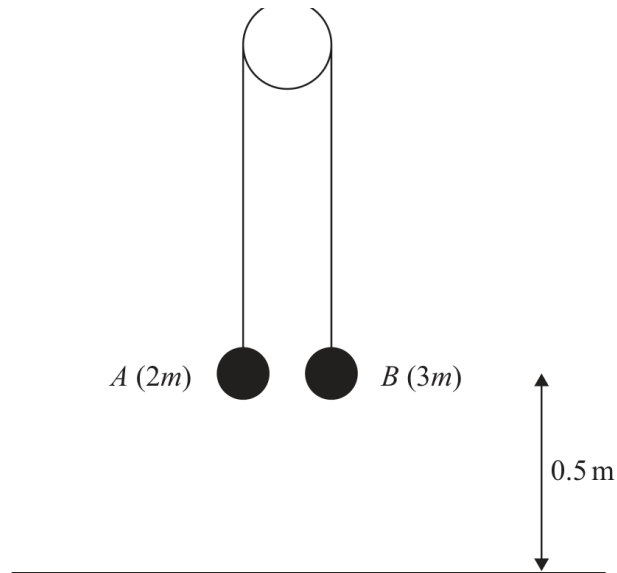
The system is released from rest and  $B$  moves downwards.

- (a) Find the tension in the string in terms of  $m$  and  $g$ .

(5)

DR ESAM AHMED

7.

**Figure 4**

The system is released from rest and  $B$  moves downwards.

- (a) Find the tension in the string in terms of  $m$  and  $g$ .

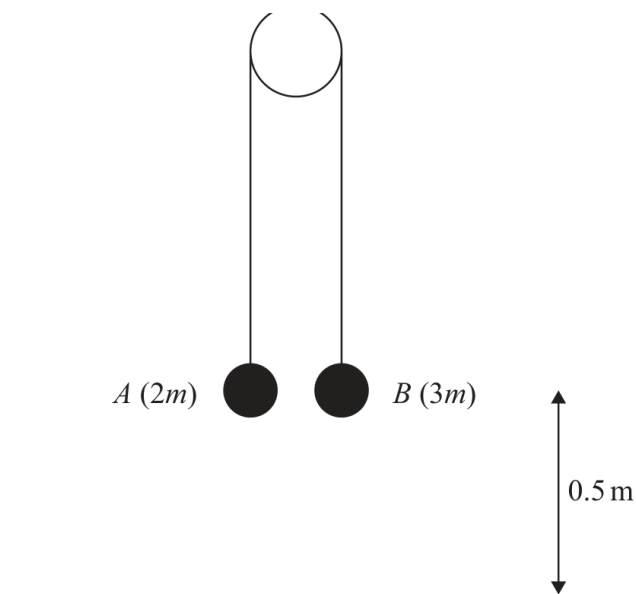
**(5)**

- (b) Find the speed of  $B$  at the instant it strikes the surface.

**(4)**

DR ESAM AHMED

7.

**Figure 4**

In the subsequent motion,  $A$  does not reach the pulley and  $B$  does not rebound after it strikes the surface.

- (c) Find the time from the instant when the system is released from rest to the instant when  $A$  first reaches a height of  $1.06\text{ m}$  above the surface.

**(6)**

Two small balls,  $P$  and  $Q$ , have masses  $kM$  and  $3M$  respectively, where  $k < 3$

The balls are attached to the ends of a light inextensible string that passes over a fixed light smooth pulley.

The system is held at rest with the string taut and the hanging parts of the string vertical, as shown in Figure 5.

The system is released from rest and, in the subsequent motion,  $P$  moves with an acceleration of magnitude  $\frac{1}{5}g$

The balls are modelled as particles.

(a) Write down an equation of motion for  $P$ .

(2)

DR ESAM AHMED

Two small balls,  $P$  and  $Q$ , have masses  $kM$  and  $3M$  respectively, where  $k < 3$

The balls are attached to the ends of a light inextensible string that passes over a fixed light smooth pulley.

The system is held at rest with the string taut and the hanging parts of the string vertical, as shown in Figure 5.

The system is released from rest and, in the subsequent motion,  $P$  moves with an acceleration of magnitude  $\frac{1}{5}g$

The balls are modelled as particles.

(b) Find the value of  $k$ .

(3)

DR ESAM AHMED

Two small balls,  $P$  and  $Q$ , have masses  $kM$  and  $3M$  respectively, where  $k < 3$

The balls are attached to the ends of a light inextensible string that passes over a fixed light smooth pulley.

The system is held at rest with the string taut and the hanging parts of the string vertical, as shown in Figure 5.

The system is released from rest and, in the subsequent motion,  $P$  moves with an acceleration of magnitude  $\frac{1}{5}g$

Given that  $M = 0.5 \text{ kg}$ ,

- (c) find the magnitude of the force exerted on the pulley by the string while  $Q$  is moving downwards.

(3)

DR ESAM AHMED

Two small balls,  $P$  and  $Q$ , have masses  $kM$  and  $3M$  respectively, where  $k < 3$

The balls are attached to the ends of a light inextensible string that passes over a fixed light smooth pulley.

The system is released from rest and, in the subsequent motion,  $P$  moves with an acceleration of magnitude  $\frac{1}{5}g$

Given that  $M = 0.5 \text{ kg}$ ,

At the instant when the system is released,  $P$  is more than  $2.5 \text{ m}$  from the pulley and  $Q$  is  $2.5 \text{ m}$  above horizontal ground.

After hitting the ground,  $Q$  rebounds with a speed of  $0.4 \text{ m s}^{-1}$

(d) Find the magnitude of the impulse received by  $Q$  when it hits the ground.

(5)

DR ESAM AHMED

Two small balls,  $P$  and  $Q$ , have masses  $kM$  and  $3M$  respectively, where  $k < 3$

The balls are attached to the ends of a light inextensible string that passes over a fixed light smooth pulley.

The system is held at rest with the string taut and the hanging parts of the string vertical, as shown in Figure 5.

The system is released from rest and, in the subsequent motion,  $P$  moves with an acceleration of magnitude  $\frac{1}{5}g$

At the instant when the system is released,  $P$  is more than 2.5 m from the pulley and  $Q$  is 2.5 m above horizontal ground.

After hitting the ground,  $Q$  rebounds with a speed of  $0.4 \text{ m s}^{-1}$

In the subsequent motion,  $P$  does not hit the pulley.

(e) Find the total time from when the balls are released until  $P$  first comes to rest.

(4)

DR ESAM AHMED

A rough plane is inclined at an angle  $\theta$  to the horizontal, where  $\tan \theta = \frac{4}{3}$

One end of a light inextensible string is attached to a package  $P$ .

The string passes over a smooth pulley that is fixed at the top of the plane.

The string from  $P$  to the pulley lies along a line of greatest slope of the plane.

The other end of the string is attached to a bucket.

The package  $P$  has mass 1 kg and is **held** at rest on the plane.

The bucket has mass 0.45 kg and hangs vertically below the pulley with the string taut, as shown in Figure 3.

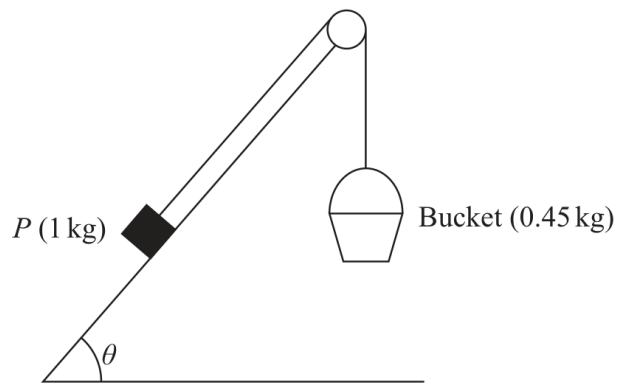
The coefficient of friction between  $P$  and the plane is  $\mu$

When a small block of mass 0.2 kg is placed in the bucket, the system is released and remains at rest in **limiting equilibrium**, with  $P$  on the point of slipping down the plane.

(a) Show that  $\mu = \frac{1}{4}$

(7)

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**Figure 3**

Additional identical blocks of mass 0.2 kg are added one at a time into the bucket until  $P$  starts to move **up** the plane.

At the instant when  $P$  starts to move up the plane, there are a **total** of  $n$  blocks in the bucket.

(b) Find

- (i) the value of  $n$
- (ii) the magnitude of the initial acceleration of  $P$

(7)

## TOPIC 07 | FAMILY 03

# Towing and coupled vehicles

**WHAT THIS FAMILY ASKS**

Use whole-system and subsystem equations for tractors, trailers, tow bars or cables.

**VARIANT ROUTE**

**V01** Whole system then one component

**V02** Known tow force used to recover system dynamics

**3 classified questions**

25 total marks

A car of mass 1200 kg is towing a trailer of mass 600 kg up a straight road, as shown in Figure 4.

The road is inclined at an angle  $\alpha$  to the horizontal, where  $\sin \alpha = \frac{1}{12}$

The driving force produced by the engine of the car is 3000 N.

The car moves with acceleration  $0.75 \text{ ms}^{-2}$

The non-gravitational resistance to motion of

- the **car** is modelled as a constant force of magnitude  $2R$  newtons
- the **trailer** is modelled as a constant force of magnitude  $R$  newtons

The car and the trailer are modelled as particles.

The tow bar between the car and trailer is modelled as a light rod that is parallel to the direction of motion.

Using the model,

(a) show that the value of  $R$  is 60

(4)

A car of mass 1200 kg is towing a trailer of mass 600 kg up a straight road, as shown in Figure 4.

The road is inclined at an angle  $\alpha$  to the horizontal, where  $\sin \alpha = \frac{1}{12}$

The driving force produced by the engine of the car is 3000 N.

The car moves with acceleration  $0.75 \text{ ms}^{-2}$

The non-gravitational resistance to motion of

- the **car** is modelled as a constant force of magnitude  $2R$  newtons
- the **trailer** is modelled as a constant force of magnitude  $R$  newtons

The car and the trailer are modelled as particles.

The tow bar between the car and trailer is modelled as a light rod that is parallel to the direction of motion.

Using the model,

(b) find the tension in the tow bar.

(3)

DR ESAM AHMED

A car of mass 1200 kg is towing a trailer of mass 600 kg up a straight road, as shown in Figure 4.

The road is inclined at an angle  $\alpha$  to the horizontal, where  $\sin \alpha = \frac{1}{12}$

The non-gravitational resistance to motion of

- the **car** is modelled as a constant force of magnitude  $2R$  newtons
- the **trailer** is modelled as a constant force of magnitude  $R$  newtons

The car and the trailer are modelled as particles.

The tow bar between the car and trailer is modelled as a light rod that is parallel to the direction of motion.

When the car and trailer are moving at a speed of  $12 \text{ ms}^{-1}$ , the tow bar breaks.

Given that the non-gravitational resistance to motion of the trailer remains unchanged,

- (c) use the model to find the further distance moved by the trailer before it first comes to rest.

(4)

DR ESAM AHMED

3. A car of mass 800 kg is towing a trailer of mass 400 kg up a straight road using a towbar. The towbar is parallel to the road and parallel to the direction of motion of the car. The road is inclined to the horizontal at an angle  $\alpha$ , where  $\sin \alpha = \frac{1}{7}$ . The engine of the car produces a constant driving force of magnitude  $D$  newtons. The resistance to the motion of the car from non-gravitational forces is modelled as a single force of magnitude 420 N. The resistance to the motion of the trailer from non-gravitational forces is modelled as a single force of magnitude 300 N. The car and trailer are modelled as particles and the towbar is modelled as a light rod.

Given that the tension in the towbar is 2060 N, find the value of  $D$ .

(7)

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3. A tractor of mass 6 tonnes is dragging a large block of mass 2 tonnes along rough horizontal ground. The cable connecting the tractor to the block is horizontal and parallel to the direction of motion.

The cable is modelled as being light and inextensible.

The driving force of the tractor is 7400 N and the resistance to the motion of the tractor is 200 N. The resistance to the motion of the block is  $R$  newtons, where  $R$  is a constant.

Given that the tension in the cable is 6000 N and the tractor is accelerating,

(a) find the value of  $R$ .

(6)

DR ESAM AHMED

3. A tractor of mass 6 tonnes is dragging a large block of mass 2 tonnes along rough horizontal ground. The cable connecting the tractor to the block is horizontal and parallel to the direction of motion.

The cable is modelled as being light and inextensible.

- (b) State how you have used the fact that the cable is modelled as being inextensible.

(1)

DR ESAM AHMED

## TOPIC 07 | FAMILY 04

# Rough horizontal dynamics

**WHAT THIS FAMILY ASKS**

Use reaction, friction and Newton's second law for motion on a rough horizontal surface, often after kinematics determines acceleration.

**VARIANT ROUTE**

**V01** Infer horizontal friction from motion data

**3 classified questions**

40 total marks

7.

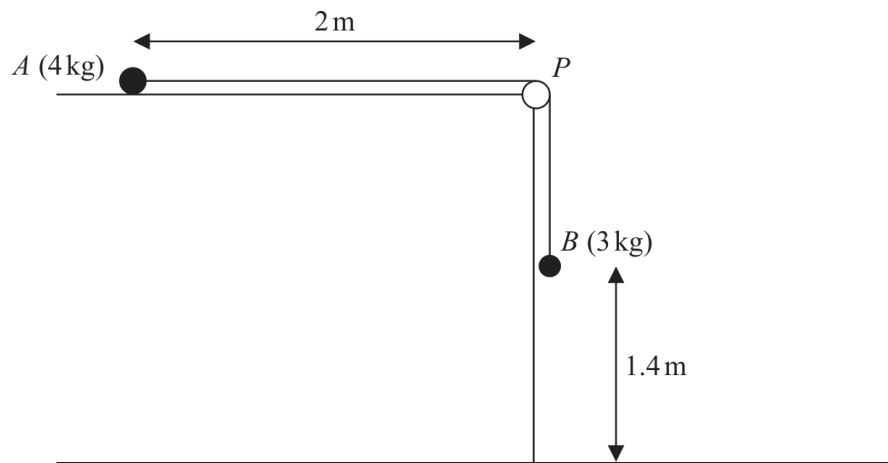


Figure 3

The system is released from rest with the string taut. At the instant of release,  $A$  is 2 m from the edge of the table and  $B$  is 1.4 m above a horizontal floor, as shown in Figure 3.

After descending with constant acceleration for 2 seconds,  $B$  hits the floor and does not rebound.

(a) Show that the acceleration of  $A$  before  $B$  hits the floor is  $0.7 \text{ m s}^{-2}$

(2)

The string is modelled as being light and inextensible and the pulley is modelled as being small, smooth and light.

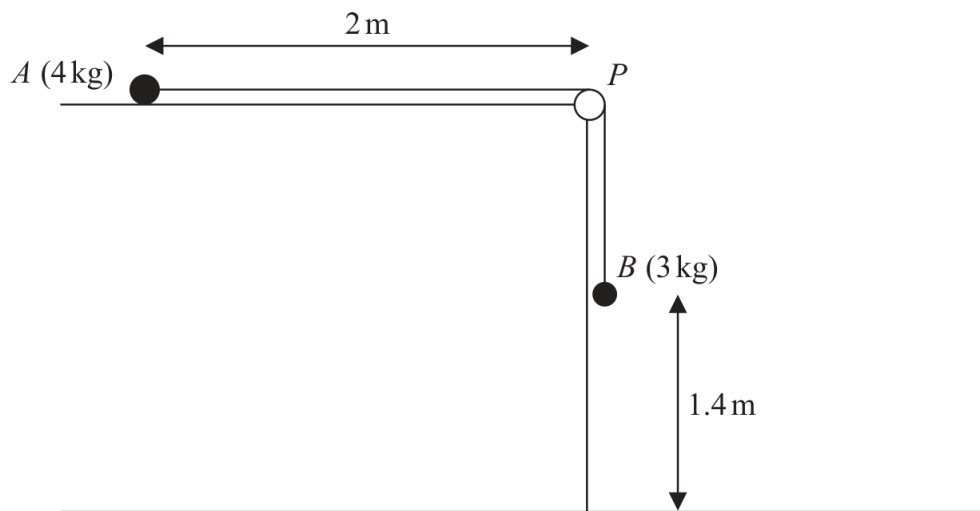
The system is released from rest with the string taut. At the instant of release,  $A$  is 2 m from the edge of the table and  $B$  is 1.4 m above a horizontal floor, as shown in Figure 3.

After descending with constant acceleration for 2 seconds,  $B$  hits the floor and does not rebound.

- (b) State which of the modelling assumptions you have used in order to answer part (a).  
(1)

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7.

**Figure 3**

The string is modelled as being light and inextensible and the pulley is modelled as being small, smooth and light.

(c) Find the magnitude of the resultant force exerted on the pulley by the string.

**(4)**

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7.

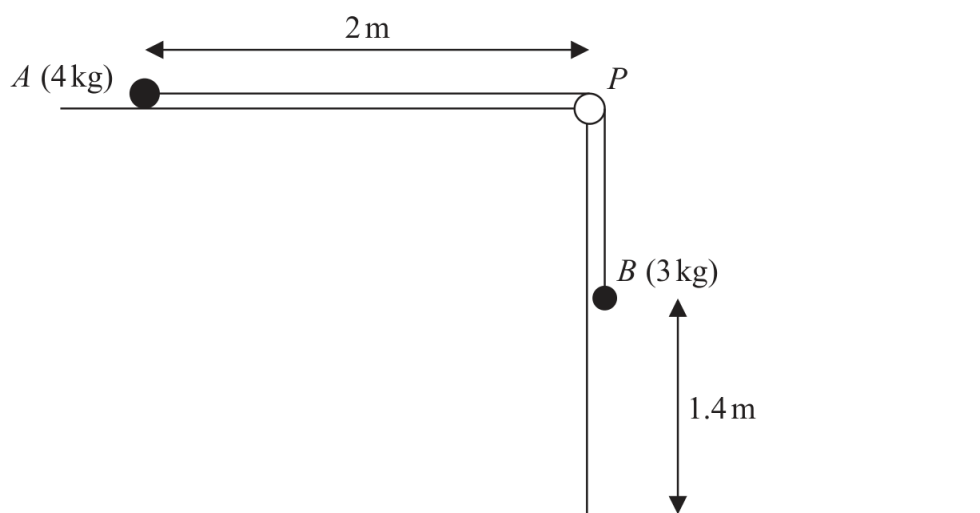


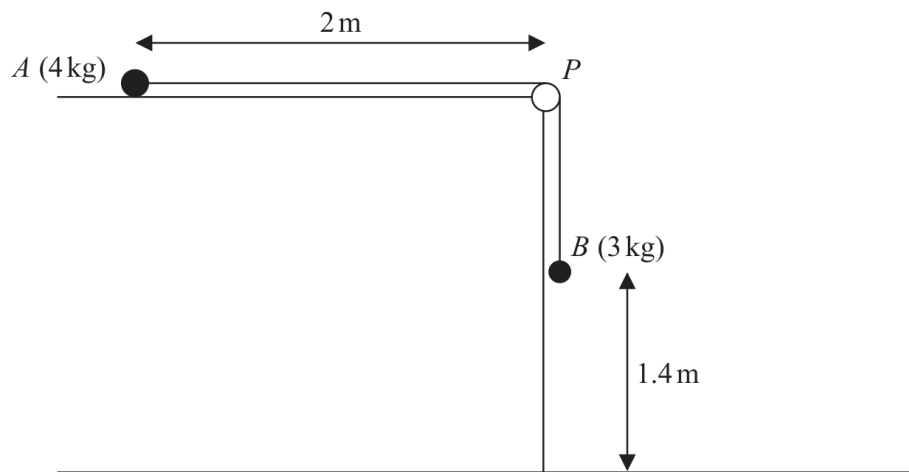
Figure 3

The coefficient of friction between  $A$  and the table is  $\mu$ .

(d) Find the value of  $\mu$ .

(6)

7.

**Figure 3**

The system is released from rest with the string taut. At the instant of release,  $A$  is 2 m from the edge of the table and  $B$  is 1.4 m above a horizontal floor, as shown in Figure 3.

After descending with constant acceleration for 2 seconds,  $B$  hits the floor and does not rebound.

(e) Determine, by calculation, whether or not  $A$  reaches the pulley.

**(5)**

2. A particle  $P$  of mass  $2m$  is moving on a rough horizontal plane when it collides directly with a particle  $Q$  of mass  $4m$  which is at rest on the plane. The speed of  $P$  immediately before the collision is  $3u$ . The speed of  $Q$  immediately after the collision is  $2u$ .

(a) Find, in terms of  $u$ , the speed of  $P$  immediately after the collision.

(3)

DR ESAM AHMED

2. A particle  $P$  of mass  $2m$  is moving on a rough horizontal plane when it collides directly with a particle  $Q$  of mass  $4m$  which is at rest on the plane. The speed of  $P$  immediately before the collision is  $3u$ . The speed of  $Q$  immediately after the collision is  $2u$ .

(b) State clearly the direction of motion of  $P$  immediately after the collision.

(1)

DR ESAM AHMED

2. A particle  $P$  of mass  $2m$  is moving on a rough horizontal plane when it collides directly with a particle  $Q$  of mass  $4m$  which is at rest on the plane. The speed of  $P$  immediately before the collision is  $3u$ . The speed of  $Q$  immediately after the collision is  $2u$ .

Following the collision,  $Q$  comes to rest after travelling a distance  $\frac{6u^2}{g}$  along the plane.

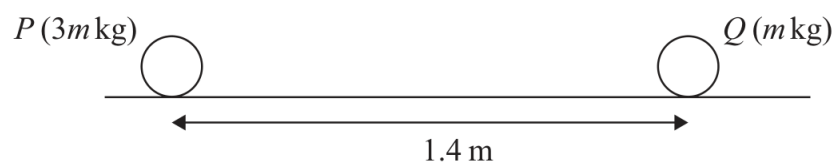
The coefficient of friction between  $Q$  and the plane is  $\mu$ .

(c) Find the value of  $\mu$ .

(6)

DR ESAM AHMED

3.

**Figure 1**

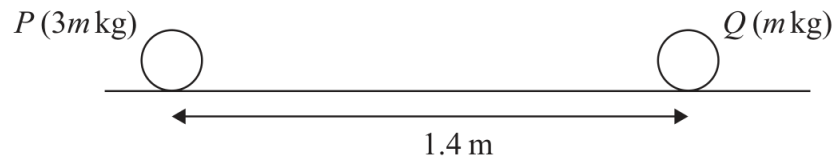
A particle  $P$  of mass  $3m \text{ kg}$  and a particle  $Q$  of mass  $m \text{ kg}$  are at rest on a rough horizontal surface. The distance between  $P$  and  $Q$  is  $1.4 \text{ m}$ , as shown in Figure 1.

An impulse of magnitude  $\lambda \text{ N s}$  is now applied to  $P$  in the direction  $PQ$ . Immediately after the impulse is applied, the speed of  $P$  is  $5 \text{ m s}^{-1}$

(a) Find  $\lambda$  in terms of  $m$ .

**(2)**

3.

**Figure 1**

A particle  $P$  of mass  $3m \text{ kg}$  and a particle  $Q$  of mass  $m \text{ kg}$  are at rest on a rough horizontal surface. The distance between  $P$  and  $Q$  is  $1.4 \text{ m}$ , as shown in Figure 1.

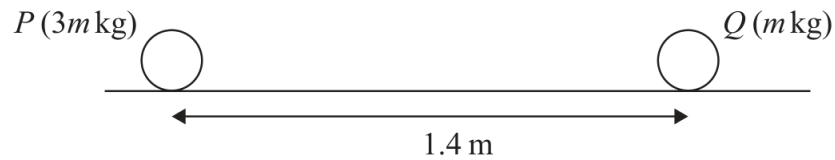
An impulse of magnitude  $\lambda \text{ N s}$  is now applied to  $P$  in the direction  $PQ$ .  
Immediately after the impulse is applied, the speed of  $P$  is  $5 \text{ m s}^{-1}$

Immediately before  $P$  collides with  $Q$ , the speed of  $P$  is  $2.5 \text{ m s}^{-1}$   
The coefficient of friction between  $P$  and the surface is  $\mu$

(b) Find the value of  $\mu$

(7)

3.

**Figure 1**

A particle  $P$  of mass  $3m \text{ kg}$  and a particle  $Q$  of mass  $m \text{ kg}$  are at rest on a rough horizontal surface. The distance between  $P$  and  $Q$  is  $1.4 \text{ m}$ , as shown in Figure 1.

Immediately before  $P$  collides with  $Q$ , the speed of  $P$  is  $2.5 \text{ m s}^{-1}$

The coefficient of friction between  $P$  and the surface is  $\mu$

Immediately after  $P$  collides with  $Q$ , the speed of  $Q$  is  $2.1 \text{ m s}^{-1}$

(c) Find the speed of  $P$  immediately after the collision.

**(3)**

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